

NAVEEN SHAJI GEORGE

Robotics Engineer

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Profile

Robotics Engineer with hands-on experience in autonomous vehicle systems, electrical architecture, embedded electronics, PCB design, CAN communication, and vehicle integration. Led Team Equinox's E&E development across two aBAJA national campaigns, combining system-level engineering, hands-on implementation, and technical leadership.

Engineering Experience

Team Equinox, Saintgits College of Engineering

Electrical & Electronics Head – aBAJA SAEINDIA

2024–2026

AIR 7 (2025) | AIR 11 (2026)

- Architected the complete electrical and electronic system of an autonomous electric Baja vehicle, spanning HV/LV power architecture, ECU architecture, CAN communication, Throttle/Brake/Steer-by-Wire, startup and safety systems, sensors, actuators, and vehicle integration.
- Designed and assembled front and rear ECU PCBs end-to-end in KiCad, covering schematics, component selection, PCB layout/routing, power regulation, CAN interfaces, sensor/actuator interfaces, MOSFET/SSR switching, watchdog protection, SMD assembly, testing, and debugging.
- Led the design and physical construction of the HV and LV wiring harnesses, integrating the AIR, 150A fuse, DC–DC converter, motor controller, ECUs, sensors, actuators, power distribution, and communication systems.
- Personally carried the electrical systems from design through physical vehicle integration, wiring, testing, and debugging of HV/LV systems, CAN, TBW/BBW/SBW, dashboard, sensors, actuators, motor controller, startup/safety systems, and Jetson Orin Nano/ROS 2 interfaces.
- Led and mentored E&E team members through design, implementation, troubleshooting, testing, and integration; authored engineering documentation and led technical presentations and inspection preparation.

Selected Projects

Hospital Service Robot

B.Tech Final Year Project

2025–2026

Robotics & Automation

- Developing an IoT-enabled hospital service robot for in-ward assistance and basic human-robot interaction, integrating sensing, control, navigation, and telemetry.

Smart CCTV Surveillance System

Intel AI Hackathon

2024

Top 25 / 300+ teams

- Developed a real-time surveillance prototype using OpenCV and Intel oneAPI for detection of weapons, masks, and hostile cues, with Telegram-based alerting.

Swarm Robot – Two-Robot Master–Slave System

B.Tech Minor Project

2024

Robotics & Automation

- Designed a two-robot master–slave system using TCP/HTTP communication, with ROS-based swarm coordination and distributed control.

Technical Skills

Programming: C++, C, Python, Bash **Robotics:** ROS 2, Gazebo, Linux

Embedded: ESP32, Arduino, Raspberry Pi, Jetson Orin Nano **Communication:** CAN, UART, I2C, SPI, TCP/IP

Electronics: KiCad, PCB Design, Power Electronics, SMD Soldering, Sensors, Actuators, Wiring Harnesses

Computer Vision: OpenCV, Basic Machine Learning **Tools:** Git, GitHub, Docker, SolidWorks, Fusion 360

Education

B.Tech. in Robotics & Automation, Saintgits College of Engineering

2022–2026 | CGPA: 6.9/10

Additional Experience & Leadership

Versius Surgical Robot Intern, Muthoot Hospitals, Kozhencherry

Jun–Jul 2025

Observed robotic-assisted surgical workflows, device checks, sterilization protocols, and peri-operative coordination using the Versius Surgical Robotic System; applied workflow and safety observations to the Hospital Service Robot project.

Other Leadership Roles: Lead Radio Jockey - GITSwave 20.22, (Saintgits),
Vice President - Anchoring Club (Saintgits.)